TUGAS RANDOM FOREST

Prediksi Pembatalan Pemesanan Hotel (Dataset Hotel Booking Demand)

Nama: _____ NIM: _____ Kelas: _____

Ringkasan hasil. Dari 10 data latih dibangun 5 pohon (masing-masing memakai 1 fitur). Untuk 1 data uji, 4 dari 5 pohon memilih **Batal (1)** dan 1 pohon memilih Tidak batal (0). Hasil voting = **Batal (1)**, sama dengan label sebenarnya. Program (11 data) menghasilkan angka yang persis sama dengan perhitungan manual.

1. Dataset dan Persiapan

Dataset: **hotel_bookings.csv** (119.390 baris, 32 kolom). Target klasifikasi: **is_canceled** (0 = tidak batal, 1 = batal). Untuk perhitungan manual dipilih 5 fitur agar mudah dihitung tangan (3 numerik, 2 kategorik, sesuai contoh di materi):

- **lead_time** (numerik) – selisih hari antara pemesanan dan kedatangan
- **adr** (numerik) – rata-rata harga kamar per hari
- **total_of_special_requests** (numerik) – jumlah permintaan khusus
- **deposit_type** (kategorik) – No Deposit / Non Refund / Refundable
- **market_segment** (kategorik) – Online TA, Offline TA/TO, Groups, dst.

Pengambilan sampel. Dengan *seed* acak tetap (=3) diambil 10 data latih (5 batal + 5 tidak batal, supaya kelasnya seimbang) dan 1 data uji. Kolom reservation_status dan reservation_status_date **tidak dipakai** karena memuat jawaban target (*data leakage*).

Data latih (10 baris)

ID	lead_time	adr	total_of_special_requests	deposit_type	market_segment	is_canceled
D1	35	145.23	0	No Deposit	Online TA	0
D2	6	161.5	2	No Deposit	Online TA	0
D3	54	75	0	No Deposit	Offline TA/TO	0
D4	44	106.47	0	No Deposit	Offline TA/TO	0
D5	240	138.6	1	No Deposit	Online TA	1
D6	263	110	0	Non Refund	Offline TA/TO	1
D7	52	88.2	0	No Deposit	Online TA	1
D8	71	120	0	Non Refund	Offline TA/TO	1
D9	66	89.1	0	No Deposit	Online TA	1
D10	15	278	2	No Deposit	Online TA	0

Data uji (1 baris)

ID	lead_time	adr	total_of_special_requests	deposit_type	market_segment	is_canceled (sebenarnya)
T1	323	62	0	Non Refund	Groups	1

2. Langkah-Langkah Perhitungan Manual

- **Bootstrap.** Untuk tiap pohon, ambil 10 data dari 10 data latih secara acak *dengan pengembalian* (satu data boleh muncul berkali-kali, ada yang tidak terpilih).
- **Pilih 1 fitur acak** untuk pohon tersebut (5 pohon memakai 5 fitur yang berbeda).
- **Gini awal** = $1 - p_0^2 - p_1^2$.
- **Kandidat split (CART biner).** Numerik: titik tengah antar dua nilai berurutan yang berbeda, uji “x ≤ t”. Kategorik: satu kategori melawan sisanya (“x = k”), sesuai contoh di materi.
- **Gini bobot** = $(n_{kiri}/n)\cdot Gini_{kiri} + (n_{kanan}/n)\cdot Gini_{kanan}$; **reduction** = Gini awal − Gini bobot.
- **Pilih split dengan reduction terbesar.** Karena hanya 1 fitur dan dibatasi kedalaman 1, tiap pohon berupa satu node akar dan dua daun (*decision stump*). Daun diberi kelas mayoritas.

- **Prediksi data uji** di setiap pohon, lalu **voting mayoritas** dari 5 pohon.

Catatan: bila dua split punya reduction sama (split cerminan pada fitur kategorik dengan 2 kategori), dipilih yang pertama; hasil prediksinya sama saja.

3. Pembentukan 5 Pohon

Pohon 1 — fitur: market_segment

(a) Sampel bootstrap: D10, D1, D7, D5, D8, D5, D8, D2, D2, D10. Data tidak terpilih (OOB): D3, D4, D6, D9.

Data	D8	D8	D10	D1	D7	D5	D5	D2	D2	D10
market_segment	Offline TA/TO	Offline TA/TO	Online TA	Online TA	Online TA	Online TA	Online TA	Online TA	Online TA	Online TA
Kelas	1	1	0	0	1	1	1	0	0	0

(b) Gini awal: kelas 0 = 5, kelas 1 = 5 → $1 - (5/10)^2 - (5/10)^2$ = **0.5000**

(c) Semua kandidat split:

Kandidat kategori (=)	Kiri n (0/1)	Gini kiri	Kanan n (0/1)	Gini kanan	Gini bobot	Reduction
Offline TA/TO	2 (0/2)	0.0000	8 (5/3)	0.4688	0.3750	0.1250
Online TA	8 (5/3)	0.4688	2 (0/2)	0.0000	0.3750	0.1250

(d) Contoh hitung split terbaik (Offline TA/TO): Gini kiri = $1 - (0/2)^2 - (2/2)^2 = 0.0000$; Gini kanan = $1 - (5/8)^2 - (3/8)^2 = 0.4688$; Gini bobot = $(2/10)\cdot 0.0000 + (8/10)\cdot 0.4688 = 0.3750$; reduction = $0.5000 - 0.3750$ = **0.1250** (terbesar → dipilih).

(e) Pohon: Akar: **market_segment = Offline TA/TO** ? Ya → **Batal (1)** | Tidak → **Tidak batal (0)**

(f) Uji T1: market_segment = Groups → market_segment = Offline TA/TO bernilai **Tidak** → prediksi pohon 1 = **Tidak batal (0)**

Pohon 2 — fitur: lead_time

(a) Sampel bootstrap: D9, D3, D4, D7, D7, D10, D10, D2, D6, D2. Data tidak terpilih (OOB): D1, D5, D8.

Data	D2	D2	D10	D10	D4	D7	D7	D3	D9	D6
lead_time	6	6	15	15	44	52	52	54	66	263
Kelas	0	0	0	0	0	1	1	0	1	1

(b) Gini awal: kelas 0 = 6, kelas 1 = 4 → $1 - (6/10)^2 - (4/10)^2$ = **0.4800**

(c) Semua kandidat split:

Kandidat ambang (≤)	Kiri n (0/1)	Gini kiri	Kanan n (0/1)	Gini kanan	Gini bobot	Reduction
10.5	2 (2/0)	0.0000	8 (4/4)	0.5000	0.4000	0.0800
29.5	4 (4/0)	0.0000	6 (2/4)	0.4444	0.2667	0.2133
48	5 (5/0)	0.0000	5 (1/4)	0.3200	0.1600	0.3200
53	7 (5/2)	0.4082	3 (1/2)	0.4444	0.4190	0.0610
60	8 (6/2)	0.3750	2 (0/2)	0.0000	0.3000	0.1800
164.5	9 (6/3)	0.4444	1 (0/1)	0.0000	0.4000	0.0800

(d) Contoh hitung split terbaik (48): Gini kiri = $1 - (5/5)^2 - (0/5)^2 = 0.0000$; Gini kanan = $1 - (1/5)^2 - (4/5)^2 = 0.3200$; Gini bobot = $(5/10)\cdot 0.0000 + (5/10)\cdot 0.3200 = 0.1600$; reduction = $0.4800 - 0.1600$ = **0.3200** (terbesar → dipilih).

(e) Pohon: Akar: **lead_time ≤ 48** ? Ya → **Tidak batal (0)** | Tidak → **Batal (1)**

(f) Uji T1: lead_time = 323 → lead_time ≤ 48 bernilai **Tidak** → prediksi pohon 2 = **Batal (1)**

Pohon 3 — fitur: total_of_special_requests

(a) Sampel bootstrap: D1, D6, D7, D7, D3, D7, D10, D9, D4, D3. Data tidak terpilih (OOB): D2, D5, D8.

Data	D1	D6	D7	D7	D3	D7	D9	D4	D3	D10
total_of_special_requests	0	0	0	0	0	0	0	0	0	2
Kelas	0	1	1	1	0	1	1	0	0	0

(b) Gini awal: kelas 0 = 5, kelas 1 = 5 → $1 - (5/10)^2 - (5/10)^2 =$ **0.5000**

(c) Semua kandidat split:

Kandidat ambang (≤)	Kiri n (0/1)	Gini kiri	Kanan n (0/1)	Gini kanan	Gini bobot	Reduction
1	9 (4/5)	0.4938	1 (1/0)	0.0000	0.4444	0.0556

(d) Contoh hitung split terbaik (1): Gini kiri = $1 - (4/9)^2 - (5/9)^2 = 0.4938$; Gini kanan = $1 - (1/1)^2 - (0/1)^2 = 0.0000$; Gini bobot = $(9/10)\cdot 0.4938 + (1/10)\cdot 0.0000 = 0.4444$; reduction = $0.5000 - 0.4444 =$ **0.0556** (terbesar → dipilih).

(e) Pohon: Akar: **total_of_special_requests ≤ 1** ? Ya → **Batal (1)** | Tidak → **Tidak batal (0)**

(f) Uji T1: total_of_special_requests = 0 → total_of_special_requests ≤ 1 bernilai **Ya** → prediksi pohon 3 = **Batal (1)**

Pohon 4 — fitur: deposit_type

(a) Sampel bootstrap: D2, D10, D9, D4, D3, D8, D9, D10, D3, D2. Data tidak terpilih (OOB): D1, D5, D6, D7.

Data	D2	D10	D9	D4	D3	D9	D10	D3	D2	D8
deposit_type	No Deposit	No Deposit	No Deposit	No Deposit	No Deposit	No Deposit	No Deposit	No Deposit	No Deposit	Non Refund
Kelas	0	0	1	0	0	1	0	0	0	1

(b) Gini awal: kelas 0 = 7, kelas 1 = 3 → $1 - (7/10)^2 - (3/10)^2 =$ **0.4200**

(c) Semua kandidat split:

Kandidat kategori (=)	Kiri n (0/1)	Gini kiri	Kanan n (0/1)	Gini kanan	Gini bobot	Reduction
No Deposit	9 (7/2)	0.3457	1 (0/1)	0.0000	0.3111	0.1089
Non Refund	1 (0/1)	0.0000	9 (7/2)	0.3457	0.3111	0.1089

(d) Contoh hitung split terbaik (No Deposit): Gini kiri = $1 - (7/9)^2 - (2/9)^2 = 0.3457$; Gini kanan = $1 - (0/1)^2 - (1/1)^2 = 0.0000$; Gini bobot = $(9/10)\cdot 0.3457 + (1/10)\cdot 0.0000 = 0.3111$; reduction = $0.4200 - 0.3111 =$ **0.1089** (terbesar → dipilih).

(e) Pohon: Akar: **deposit_type = No Deposit** ? Ya → **Tidak batal (0)** | Tidak → **Batal (1)**

(f) Uji T1: deposit_type = Non Refund → deposit_type = No Deposit bernilai **Tidak** → prediksi pohon 4 = **Batal (1)**

Pohon 5 — fitur: adr

(a) Sampel bootstrap: D9, D2, D9, D1, D5, D6, D5, D5, D4, D4. Data tidak terpilih (OOB): D3, D7, D8, D10.

Data	D9	D9	D4	D4	D6	D5	D5	D5	D1	D2
adr	89.1	89.1	106.47	106.47	110	138.6	138.6	138.6	145.23	161.5
Kelas	1	1	0	0	1	1	1	1	0	0

(b) Gini awal: kelas 0 = 4, kelas 1 = 6 → $1 - (4/10)^2 - (6/10)^2 =$ **0.4800**

(c) Semua kandidat split:

Kandidat ambang (≤)	Kiri n (0/1)	Gini kiri	Kanan n (0/1)	Gini kanan	Gini bobot	Reduction
97.785	2 (0/2)	0.0000	8 (4/4)	0.5000	0.4000	0.0800
108.235	4 (2/2)	0.5000	6 (2/4)	0.4444	0.4667	0.0133
124.3	5 (2/3)	0.4800	5 (2/3)	0.4800	0.4800	0.0000
141.915	8 (2/6)	0.3750	2 (2/0)	0.0000	0.3000	0.1800
153.365	9 (3/6)	0.4444	1 (1/0)	0.0000	0.4000	0.0800

(d) Contoh hitung split terbaik (141.915): Gini kiri = $1 - (2/8)^2 - (6/8)^2 = 0.3750$; Gini kanan = $1 - (2/2)^2 - (0/2)^2 = 0.0000$; Gini bobot = $(8/10)\cdot 0.3750 + (2/10)\cdot 0.0000 = 0.3000$; reduction = $0.4800 - 0.3000 =$ **0.1800** (terbesar → dipilih).

(e) Pohon: Akar: **adr ≤ 141.915** ? Ya → **Batal (1)** | Tidak → **Tidak batal (0)**

(f) Uji T1: adr = 62 → adr ≤ 141.915 bernilai **Ya** → prediksi pohon 5 = **Batal (1)**

4. Prediksi Data Uji dan Voting

Pohon	Fitur	Aturan	Nilai T1	Prediksi
1	market_segment	market_segment = Offline TA/TO	Groups	0
2	lead_time	lead_time ≤ 48	323	1
3	total_of_special_requests	total_of_special_requests ≤ 1	0	1
4	deposit_type	deposit_type = No Deposit	Non Refund	1
5	adr	adr ≤ 141.915	62	1

Voting: kelas 0 (tidak batal) = 1 suara, kelas 1 (batal) = 4 suara → **prediksi akhir = 1 (Batal)**. Label sebenarnya = 1 (Batal) → **prediksi BENAR**.

5. Program

Link program (Google Colab / GitHub – pastikan dapat diakses):

LINK: _____

Kode lengkap juga dilampirkan pada Lampiran. Program terdiri atas dua bagian:

- **Bagian A – 11 data**: Random Forest dibuat dari nol (bootstrap, 1 fitur per pohon, Gini, voting). Angkanya persis sama dengan perhitungan manual di atas.
- **Bagian B – seluruh data** (119.390 baris): *RandomForestClassifier* scikit-learn seperti biasa, dengan pembagian 80% latih / 20% uji.

5.1 Keluaran Bagian A (11 data)

```
Ukuran dataset: (119390, 32)
DATA LATIH:
      lead_time     adr  total_of_special_requests deposit_type market_segment  is_canceled
D1           35  145.23                          0   No Deposit      Online TA            0
D2            6  161.50                          2   No Deposit      Online TA            0
D3           54   75.00                          0   No Deposit  Offline TA/TO            0
D4           44  106.47                          0   No Deposit  Offline TA/TO            0
D5          240  138.60                          1   No Deposit      Online TA            1
D6          263  110.00                          0   Non Refund  Offline TA/TO            1
D7           52   88.20                          0   No Deposit      Online TA            1
D8           71  120.00                          0   Non Refund  Offline TA/TO            1
D9           66   89.10                          0   No Deposit      Online TA            1
D10          15  278.00                          2   No Deposit      Online TA            0

DATA UJI:
  lead_time  adr  total_of_special_requests deposit_type market_segment  is_canceled
       323 62.0                          0   Non Refund         Groups            1

Pohon 1 | fitur = market_segment | bootstrap = ['D10', 'D1', 'D7', 'D5', 'D8', 'D5', 'D8', 'D2', 'D2', 'D10']
  Gini awal = 0.5000
  kandidat  Offline TA/TO | kiri(0/1)=0/2 kanan(0/1)=5/3 | Gini bobot=0.3750 | reduction=0.1250
  kandidat      Online TA | kiri(0/1)=5/3 kanan(0/1)=0/2 | Gini bobot=0.3750 | reduction=0.1250
  -> Split terbaik: market_segment == Offline TA/TO  | kiri -> 1, kanan -> 0

Pohon 2 | fitur = lead_time | bootstrap = ['D9', 'D3', 'D4', 'D7', 'D7', 'D10', 'D10', 'D2', 'D6', 'D2']
  Gini awal = 0.4800
  kandidat           10.5 | kiri(0/1)=2/0 kanan(0/1)=4/4 | Gini bobot=0.4000 | reduction=0.0800
  kandidat           29.5 | kiri(0/1)=4/0 kanan(0/1)=2/4 | Gini bobot=0.2667 | reduction=0.2133
  kandidat           48.0 | kiri(0/1)=5/0 kanan(0/1)=1/4 | Gini bobot=0.1600 | reduction=0.3200
  kandidat           53.0 | kiri(0/1)=5/2 kanan(0/1)=1/2 | Gini bobot=0.4190 | reduction=0.0610
  kandidat           60.0 | kiri(0/1)=6/2 kanan(0/1)=0/2 | Gini bobot=0.3000 | reduction=0.1800
  kandidat          164.5 | kiri(0/1)=6/3 kanan(0/1)=0/1 | Gini bobot=0.4000 | reduction=0.0800
  -> Split terbaik: lead_time <= 48.0  | kiri -> 0, kanan -> 1

Pohon 3 | fitur = total_of_special_requests | bootstrap = ['D1', 'D6', 'D7', 'D7', 'D3', 'D7', 'D10', 'D9', 'D4', 'D3']
  Gini awal = 0.5000
  kandidat            1.0 | kiri(0/1)=4/5 kanan(0/1)=1/0 | Gini bobot=0.4444 | reduction=0.0556
  -> Split terbaik: total_of_special_requests <= 1.0  | kiri -> 1, kanan -> 0

Pohon 4 | fitur = deposit_type | bootstrap = ['D2', 'D10', 'D9', 'D4', 'D3', 'D8', 'D9', 'D10', 'D3', 'D2']
  Gini awal = 0.4200
  kandidat     No Deposit | kiri(0/1)=7/2 kanan(0/1)=0/1 | Gini bobot=0.3111 | reduction=0.1089
  kandidat     Non Refund | kiri(0/1)=0/1 kanan(0/1)=7/2 | Gini bobot=0.3111 | reduction=0.1089
  -> Split terbaik: deposit_type == No Deposit  | kiri -> 0, kanan -> 1

Pohon 5 | fitur = adr | bootstrap = ['D9', 'D2', 'D9', 'D1', 'D5', 'D6', 'D5', 'D5', 'D4', 'D4']
  Gini awal = 0.4800
  kandidat         97.785 | kiri(0/1)=0/2 kanan(0/1)=4/4 | Gini bobot=0.4000 | reduction=0.0800
  kandidat        108.235 | kiri(0/1)=2/2 kanan(0/1)=2/4 | Gini bobot=0.4667 | reduction=0.0133
  kandidat          124.3 | kiri(0/1)=2/3 kanan(0/1)=2/3 | Gini bobot=0.4800 | reduction=0.0000
  kandidat        141.915 | kiri(0/1)=2/6 kanan(0/1)=2/0 | Gini bobot=0.3000 | reduction=0.1800
  kandidat        153.365 | kiri(0/1)=3/6 kanan(0/1)=1/0 | Gini bobot=0.4000 | reduction=0.0800
  -> Split terbaik: adr <= 141.915  | kiri -> 1, kanan -> 0

Prediksi tiap pohon : [0, 1, 1, 1, 1]
Voting              : 0 (tidak batal) = 1 | 1 (batal) = 4
Prediksi akhir      : 1 (Batal)
Label sebenarnya    : 1
```

5.2 Keluaran Bagian B (seluruh data)

```
BAGIAN B - seluruh data | latih: (95512, 29) | uji: (23878, 29)

[Random Forest standar: 100 pohon]
  Accuracy  = 0.8942
  Precision = 0.8871
  Recall    = 0.8187
  F1-score  = 0.8515
  Confusion matrix [[TN FP],[FN TP]]:
 [[14111   922]
 [ 1604  7241]]

[Skema tugas: 5 pohon, max_features=1, max_depth=1]
  Accuracy  = 0.7245
  Precision = 0.9935
  Recall    = 0.2580
  F1-score  = 0.4096
  Confusion matrix [[TN FP],[FN TP]]:
 [[15018    15]
 [ 6563  2282]]

10 fitur terpenting (model 100 pohon):
 deposit_type                0.1253
lead_time                    0.1221
country                      0.1132
adr                          0.0732
total_of_special_requests    0.0560
arrival_date_day_of_month    0.0526
agent                        0.0517
arrival_date_week_number     0.0496
market_segment               0.0476
stays_in_week_nights         0.0342
```

Interpretasi

- **Random Forest standar (100 pohon, semua fitur)**: accuracy 0,8942, F1 0,8515 — model bekerja baik pada data uji 23.878 baris.
- **Skema tugas (5 pohon, 1 fitur, kedalaman 1)**: accuracy hanya 0,7245 dan recall 0,2580. Pohon yang sangat dangkal terlalu sederhana (*underfitting*) sehingga banyak pembatalan terlewat. Ini menunjukkan alasan Random Forest nyata memakai pohon yang dalam dan jumlah pohon yang banyak.
- Fitur terpenting: deposit_type, lead_time, country, adr, dan total_of_special_requests — sejalan dengan fitur yang dipakai pada contoh manual.

Lampiran: Kode Program Lengkap (Python)

```
# ======================================================================
# TUGAS RANDOM FOREST - Dataset: Hotel Booking Demand (hotel_bookings.csv)
# Target : is_canceled  (0 = tidak batal, 1 = batal)
# BAGIAN A : 11 data (10 latih + 1 uji), 5 pohon, 1 fitur/pohon (dibuat dari nol,
#            hasilnya identik dengan perhitungan manual)
# BAGIAN B : seluruh data, Random Forest "seperti biasa" (scikit-learn)
# ======================================================================
import numpy as np
import pandas as pd
from collections import Counter

CSV_PATH = "hotel_bookings.csv"      # ubah jika lokasi file berbeda
df = pd.read_csv(CSV_PATH)
print("Ukuran dataset:", df.shape)

# ----------------------------------------------------------------------
# BAGIAN A  -  RANDOM FOREST DARI NOL PADA 11 DATA
# ----------------------------------------------------------------------
FEATS   = ["lead_time", "adr", "total_of_special_requests", "deposit_type", "market_segment"]
NUMERIC = {"lead_time", "adr", "total_of_special_requests"}
TARGET  = "is_canceled"

# 1) Ambil 10 data latih (5 batal + 5 tidak batal) dan 1 data uji (batal)
rs = np.random.RandomState(3)
c1 = df[df[TARGET] == 1].sample(6, random_state=rs)
c0 = df[df[TARGET] == 0].sample(5, random_state=rs)
train = pd.concat([c1.iloc[:5], c0]).sample(frac=1, random_state=rs)[FEATS + [TARGET]]
test  = c1.iloc[[5]][FEATS + [TARGET]]
train = train.reset_index(drop=True)
train.index = [f"D{i+1}" for i in range(10)]
print("\nDATA LATIH:\n", train.to_string())
print("\nDATA UJI:\n", test.to_string(index=False))

# 2) Fungsi Gini & pencarian split terbaik (CART biner)
def gini(y):
    n = len(y)
    if n == 0:
        return 0.0
    return 1 - sum((v / n) ** 2 for v in Counter(y).values())

def evaluate_splits(x, y, feat):
    """Numerik: kandidat = titik tengah antar nilai berurutan (x <= t).
       Kategorik: kandidat = satu kategori vs sisanya (x == k)."""
    x, y = np.asarray(x), np.asarray(y)
    vals = sorted(set(x))
    cands = [(vals[i] + vals[i + 1]) / 2 for i in range(len(vals) - 1)] if feat in NUMERIC else vals
    g0, rows = gini(y), []
    for c in cands:
        m = (x <= c) if feat in NUMERIC else (x == c)
        yl, yr = y[m], y[~m]
        w = len(yl) / len(y) * gini(yl) + len(yr) / len(y) * gini(yr)
        rows.append(dict(cand=c, nL=len(yl), L0=int((yl == 0).sum()), L1=int((yl == 1).sum()),
                         nR=len(yr), R0=int((yr == 0).sum()), R1=int((yr == 1).sum()),
                         gini_w=w, reduction=g0 - w))
    return g0, rows

def majority(n0, n1, tie=1):
    return tie if n0 == n1 else int(n1 > n0)

# 3) Bangun 5 pohon: tiap pohon = bootstrap 10 data + 1 fitur acak (tanpa pengulangan fitur)
rng  = np.random.RandomState(17)
perm = rng.permutation(len(FEATS))
forest = []
for t in range(5):
    boot = rng.randint(0, 10, 10)                # bootstrap (dengan pengembalian)
    feat = FEATS[perm[t]]
    sub  = train.iloc[boot]
    g0, rows = evaluate_splits(sub[feat].values, sub[TARGET].values, feat)
    best = max(rows, key=lambda r: r["reduction"])
    forest.append(dict(feat=feat, cand=best["cand"],
                       left=majority(best["L0"], best["L1"]),
                       right=majority(best["R0"], best["R1"])))
    print(f"\nPohon {t+1} | fitur = {feat} | bootstrap = {[f'D{i+1}' for i in boot]}")
    print(f"  Gini awal = {g0:.4f}")
    for r in rows:
        print(f"  kandidat {str(r['cand']):>14} | kiri(0/1)={r['L0']}/{r['L1']} kanan(0/1)={r['R0']}/{r['R1']}"
              f" | Gini bobot={r['gini_w']:.4f} | reduction={r['reduction']:.4f}")
    op = "<=" if feat in NUMERIC else "=="
    print(f"  -> Split terbaik: {feat} {op} {best['cand']}  | kiri -> {forest[-1]['left']}, kanan -> {forest[-1]['right']}")

# 4) Prediksi data uji + voting
def predict_tree(tr, x):
    go_left = (x <= tr["cand"]) if tr["feat"] in NUMERIC else (x == tr["cand"])
    return tr["left"] if go_left else tr["right"]

votes = [predict_tree(tr, test.iloc[0][tr["feat"]]) for tr in forest]
final = Counter(votes).most_common(1)[0][0]
print("\nPrediksi tiap pohon :", votes)
print("Voting              : 0 (tidak batal) =", votes.count(0), "| 1 (batal) =", votes.count(1))
print("Prediksi akhir      :", final, "(Batal)" if final == 1 else "(Tidak batal)")
print("Label sebenarnya    :", int(test.iloc[0][TARGET]))

# ----------------------------------------------------------------------
# BAGIAN B  -  SELURUH DATA (seperti biasa, scikit-learn)
# ----------------------------------------------------------------------
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OrdinalEncoder
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
```

```
data = df.copy()
# reservation_status & reservation_status_date = kebocoran target (data leakage) -> dibuang
data = data.drop(columns=["reservation_status", "reservation_status_date"])
# penanganan nilai kosong
data["children"] = data["children"].fillna(0)
data["country"]  = data["country"].fillna("Unknown")
data[["agent", "company"]] = data[["agent", "company"]].fillna(0)

X, y = data.drop(columns=[TARGET]), data[TARGET]
cat_cols = [c for c in X.columns if not pd.api.types.is_numeric_dtype(X[c])]
X[cat_cols] = OrdinalEncoder().fit_transform(X[cat_cols])      # kategorik -> angka

X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
print("\n\nBAGIAN B - seluruh data | latih:", X_tr.shape, "| uji:", X_te.shape)

def laporan(nama, model):
    model.fit(X_tr, y_tr)
    p = model.predict(X_te)
    print(f"\n[{nama}]")
    print(f"  Accuracy  = {accuracy_score(y_te, p):.4f}")
    print(f"  Precision = {precision_score(y_te, p):.4f}")
    print(f"  Recall    = {recall_score(y_te, p):.4f}")
    print(f"  F1-score  = {f1_score(y_te, p):.4f}")
    print("  Confusion matrix [[TN FP],[FN TP]]:\n", confusion_matrix(y_te, p))
    return model

# (i) model standar
rf = laporan("Random Forest standar: 100 pohon", RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1))
# (ii) skema tugas: 5 pohon, 1 fitur per split, kedalaman 1
laporan("Skema tugas: 5 pohon, max_features=1, max_depth=1",
        RandomForestClassifier(n_estimators=5, max_features=1, max_depth=1, random_state=42, n_jobs=-1))

imp = pd.Series(rf.feature_importances_, index=X.columns).sort_values(ascending=False).head(10)
print("\n10 fitur terpenting (model 100 pohon):\n", imp.round(4).to_string())
```